

1 Solution — GIAM 6.2

Problem 3

1.1 Problem

Explain why both of the relations just described (in problems 1 and 2) have the transitive property.

1.2 Setup

Recall the relations:

- $S = \{(x, y) : x \text{ is smarter than } y\}$, on a population P .
- $A = \{(x, y) : x \text{ has the same astrological sign as } y\}$, on the same P .

A relation R on a set X is **transitive** if, for all $x, y, z \in X$,

$$(x, y) \in R \text{ and } (y, z) \in R \implies (x, z) \in R.$$

1.3 S Is Transitive

Suppose $(x, y) \in S$ and $(y, z) \in S$ — that is, x is smarter than y , and y is smarter than z . Following the strict reading from Problem 1, “smarter than” behaves like $>$: people can be compared on a single intelligence scale, and “ x is smarter than y ” says x ranks above y . Two such inequalities chain in the obvious way:

$$x \text{ above } y \text{ and } y \text{ above } z \implies x \text{ above } z,$$

so x is smarter than z , i.e. $(x, z) \in S$.

Formally: let $\iota : P \rightarrow \mathbb{R}$ rank the population by intelligence (or, more cautiously, take any strict order on P that captures the everyday relation). Then

$$(x, y) \in S \iff \iota(x) > \iota(y),$$

and transitivity of $>$ on $\mathbb{R}$ gives transitivity of S at once.

1.4 A Is Transitive

Suppose $(x, y) \in A$ and $(y, z) \in A$ — that is, x has the same sign as y , and y has the same sign as z . Then x has the same sign as z : anyone matching y matches anyone else matching y .

Formally: with $\sigma : P \rightarrow Z$ the sign function from Problem 2,

$$(x, y) \in A \iff \sigma(x) = \sigma(y),$$

and equality on Z is transitive:

$$\sigma(x) = \sigma(y) \text{ and } \sigma(y) = \sigma(z) \implies \sigma(x) = \sigma(z),$$

so $(x, z) \in A$.

1.5 Remark — Both Pull Back from Transitive Targets

The two arguments are the same proof in different costumes. Given a function $f : X \rightarrow Y$ and a relation Q on Y , define

$$R = \{(x, y) \in X \times X : (f(x), f(y)) \in Q\}.$$

If Q is transitive on Y , so is R on X : from $(f(x), f(y)) \in Q$ and $(f(y), f(z)) \in Q$, transitivity of Q gives $(f(x), f(z)) \in Q$, hence $(x, z) \in R$.

Our two relations instantiate this recipe:

- S : $f = \iota$ (intelligence rank, $P \rightarrow \mathbb{R}$); Q is $>$ on $\mathbb{R}$ — a strict order, transitive.
- A : $f = \sigma$ (zodiac sign, $P \rightarrow Z$); Q is $=$ on Z — equality, transitive.

The earlier differences (asymmetry of S , symmetry of A) trace back to analogous differences between $>$ and $=$ on the target. Transitivity is shared because both targets are transitive.